

Order of Magnitude

Spot quiz · 10 marks total

Name: _____

Date: _____

1. A bacterium is about $3\text{ }\mu\text{m}$ across and a red blood cell is about $30\text{ }\mu\text{m}$ across. Calculate how many times bigger the red blood cell is, and state this as an order of magnitude.

[2 marks]

2. A plant cell is $5 \times 10^{-5}\text{ m}$ wide and a bacterium is $5 \times 10^{-6}\text{ m}$ wide. Calculate how many times wider the plant cell is.

[2 marks]

3. A virus is $2 \times 10^{-8}\text{ m}$ across and a bacterium is $2 \times 10^{-6}\text{ m}$ across. Calculate how many times bigger the bacterium is than the virus, and state this as an order of magnitude.

[2 marks]

4. An animal cell is $40\text{ }\mu\text{m}$ across and a mitochondrion inside it is $4\text{ }\mu\text{m}$ across. Calculate how many times bigger the animal cell is.

[2 marks]

5. A cell nucleus is $1 \times 10^{-5}\text{ m}$ across and a ribosome is $1 \times 10^{-7}\text{ m}$ across. Calculate how many orders of magnitude bigger the nucleus is.

[2 marks]